

Liquid stability, LLE and VLLE

After midterm · Session 3 of 6 · 180 minutes

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The question for today

When does a calculated liquid split represent stable equilibrium?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Phase equilibrium includes stability

Equality of chemical potentials identifies stationary candidate states. Stable equilibrium requires the lowest admissible total Gibbs energy.

$$g_{mix}/RT = \sum_i x_i \ln x_i + g^E/(RT)$$

A split can lower G even when a homogeneous liquid is locally stable.

The common-tangent conditions

$$g'(x^\alpha) = g'(x^\beta) = \frac{g(x^\beta) - g(x^\alpha)}{x^\beta - x^\alpha}$$

The tangent must not lie above $g(x)$ anywhere in the admitted composition interval.

A trivial root $x^\alpha = x^\beta$ does not establish two distinct liquid phases.

A lever-rule calculation

Synthetic tie line: $x_1^\alpha=0.2$, $x_1^\beta=0.8$. Feed $z_1=0.5$.

$$f_\beta = \frac{z_1 - x_1^\alpha}{x_1^\beta - x_1^\alpha} = 0.5$$

For $z_1=0.35$, $f^\beta=0.25$. At the same coexistence conditions, the phase amounts change but the tie-line endpoints do not.

An independent check

Use the calculated endpoints to close the component balance independently. Check that the tangent supports the entire admitted Gibbs curve.

One demonstration • 20 minutes

Open the core lab

In Lab 04 select a supported liquid split and save it. Change only overall composition inside the tie line. Compare phase compositions with phase fractions.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 04, retain one supported tie line and compare two overall compositions inside it. Verify the lever rule by hand and explain the global stability check.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

When does a calculated liquid split represent stable equilibrium?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Follow-up practice • 30–40 minutes

In Lab 04, retain one supported tie line and compare two overall compositions inside it. Verify the lever rule by hand and explain the global stability check.

Assessment 3 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is included. With pre-class reading, independent work totals 45–60 minutes.

Optional study

Choose one extension: NRTL tie-line fitting in Lab 05, paired bootstrap from independent replicates, or feasible three-phase amounts in Lab 06.

Optional paper bridge: recover partial molar properties from tangent intercepts.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.